Multi-target IC v3: k-term table (ic5), 22-32 bit groups

Multi-target IC v3 results: the k-term table (ic5) is cheaper than the strongest rho in every cell

Protocol: `PROTOCOL.md` (frozen in f71b42cc4f; amendment A1 after the rho-restart fix 75dbc70b36, before any holdout result). ic5 parameters frozen from the tune set in `frozen_ic5.json` (mu = 1.0, k = 4) before the holdout. Holdout: 44 fresh prime-order curves (indices 1..11 at 22, 24, 28, 32 bits), seeds 21 and 22, T in {1024, 4096, 16384}: **4488 runs, 0 errors, every target of every run verified**. Data: `results/holdout/rows.jsonl` (artifact store), `results/holdout/runs.json` (compact counters in Git), `analysis.txt`, `summary.json`, `noverify.txt`. Every number here is [pcc:unverified] until the claim is signed.

Cost per target in GAE = [(c_I I + M + S)/(c_I + 3) + h (H + L)] / T. Each ratio is the median per-configuration ratio (22 configurations per cell); ranges are over c_I in {3, 10, 30} and h in {0, 0.1, 1}. rhobest = per configuration and price, the cheapest of 12 rho runs (rho, rho2, rho2x, rhoseq at kappa 0.5, 0.7, 1.0) after the restart fix; bsgs2 = best of three mf.

Main table

bits	T	ic5/rhobest	ic5/bsgs2	ic5/ic4	ic4/rhobest	ic4 vs rhobest (c_I 3/10/30)
22	1024	0.919-0.963	1.275-1.361	0.830-0.894	1.071-1.130	dearer / dearer / dearer
22	4096	0.931-0.972	1.181-1.278	0.927-0.970	0.984-1.016	unranked / unranked / unranked
22	16384	0.931-0.982	1.112-1.197	0.969-0.992	0.960-0.990	cheaper / cheaper / cheaper
24	1024	0.915-0.975	1.344-1.405	0.786-0.840	1.125-1.213	dearer / dearer / dearer
24	4096	0.912-0.962	1.243-1.334	0.902-0.944	0.987-1.042	unranked / dearer / dearer
24	16384	0.925-0.968	1.158-1.255	0.958-0.984	0.954-0.988	cheaper / cheaper / cheaper
28	1024	0.875-0.959	1.430-1.447	0.634-0.662	1.360-1.502	dearer / dearer / dearer
28	4096	0.870-0.946	1.367-1.410	0.831-0.863	1.037-1.134	dearer / dearer / dearer
28	16384	0.880-0.945	1.283-1.358	0.922-0.949	0.944-1.019	cheaper / unranked / unranked
32	1024	0.874-0.975	1.469-1.491	0.473-0.483	1.826-2.060	dearer / dearer / dearer
32	4096	0.854-0.952	1.423-1.432	0.707-0.723	1.190-1.338	dearer / dearer / dearer
32	16384	0.848-0.942	1.374-1.402	0.862-0.879	0.974-1.090	unranked / dearer / dearer

Section 5a decisions (bootstrap 95% interval of the median ratio at all three h):

- **ic5 vs rhobest: cheaper in all 36 (cell, c_I)**, from 22 to 32 bits and T = 1024 to 16384.
- ic5 vs ic4: cheaper in all 36. ic5 vs bsgs2: dearer in all 36 (1.11x to 1.49x).
- ic4 vs rhobest: cheaper only at T = 16384 for 22 and 24 b (and 28 b at c_I = 3); dearer at every T = 1024 cell and from 28 b at T = 4096; at 32 b, T = 16384 unranked or dearer.
- ic4 vs bsgs2: dearer in all 36.

Without the verification phase that every arm shares (about 30 to 50 GAE per target), at I = 10 M (`noverify.txt`): ic5/rhobest 0.883-0.962, ic5/bsgs2 1.42-1.53, ic4/rhobest 0.92-2.04.

Why ic5 wins where ic4 did not (counted phase split, `rows.jsonl` phases)

ic4 needs about 2m relations for m ~ sqrt(table) unknown factor-base logs. ic5 with k = 4 reaches the same table size from m ~ table^(1/4) logs: at 32 b, T = 1024 it has m = 48 unknowns and needs about 61 relations, against ic4's m = 1218 and 2436 relations, so its relation phase is a few per cent of the run. What is left is the table (one shared inversion per two x-classes) and the paired target trials, and the measured trials per target match n/(2K) (32 b, T = 1024: 1396 measured vs 1380 predicted with K = 1 555 415 classes). ic5's advantage over ic4 therefore grows with the group and shrinks with T (0.47 at 32 b, T = 1024; 0.86-0.99 at T = 16384).

Against rho the remaining gain is constant-factor: x-only classes from shared inversions in both the table and the target trials, against rho's walk steps. It does not change the exponent, and BSGS, which tiles its windows, stays 1.1x to 1.5x cheaper (`../mt_bsgs_gap_20261111/README.md`).

Pre-registered predictions vs outcome

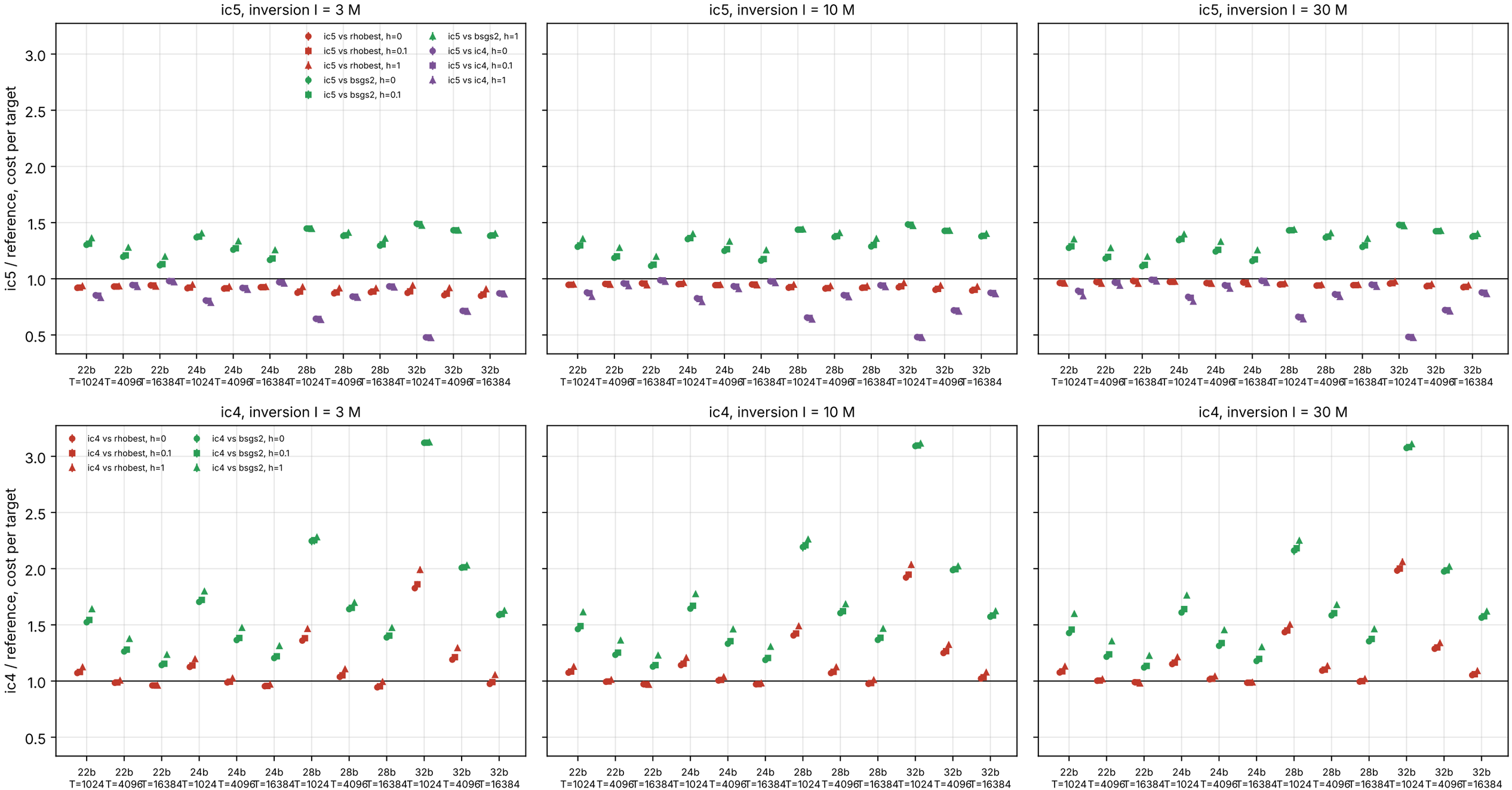
prediction	outcome
ic5/ic4 < 1 everywhere, largest gain at T = 1024 (about 0.85)	holds; the gain is larger than predicted and grows with size (0.47 at 32 b)
ic5/rhobest cheaper at T = 4096 and 16384 at every size	holds
ic5/rhobest at T = 1024 about 0.92, cheaper or unranked	cheaper in every size (0.87-0.98)
ic4/rhobest at 28 and 32 b: cheaper at 16384, dearer at 1024	dearer at 1024 holds; at 16384 only 28 b at c_I = 3 is cheaper, 32 b is unranked or dearer (wrong)
both dearer than bsgs2 everywhere	holds

Answers to the questions

1. (option 2, small T) Yes: ic5 moves the crossover against the strongest rho down to T = 1024 at every size tested.
2. (option 1, larger groups) ic4's v2 pattern does not hold at 28 and 32 b: its relation phase grows like sqrt of the table, and it falls behind rho except at T = 16384. ic5 holds or improves its margin with size (the largest gains, 0.85-0.88, are at 28 and 32 b).

Limits

Groups of 2^22 to 2^33; costs counted in field operations, not timed; memory unbounded for every arm; one local host (this container has no AWS identity, so not isolab-eks); the rho family is this harness's, not ecbench's, and was fixed during this study (`../mt_rho_start_fix_20261111/`). The comparison is per target in a batch of T targets on one curve, the regime in which multi-target methods apply; it says nothing about single-target ECDLP. The ic5 tune set chose k = 4, the largest k allowed; k = 5 was not tried.


ic5, inversion I = 3 M
ic5, inversion I = 10 M
ic5, inversion I = 30 M
ic5 / reference, cost per target
ic5 vs rhobest, h=0
ic5 vs rhobest, h=0.1
ic5 vs rhobest, h=1
ic5 vs bsgs2, h=0
ic5 vs bsgs2, h=0.1
ic5 vs bsgs2, h=1
ic5 vs ic4, h=0
ic5 vs ic4, h=0.1
ic5 vs ic4, h=1
ic4, inversion I = 3 M
ic4, inversion I = 10 M
ic4, inversion I = 30 M
ic4 / reference, cost per target
ic4 vs rhobest, h=0
ic4 vs rhobest, h=0.1
ic4 vs rhobest, h=1
ic4 vs bsgs2, h=0
ic4 vs bsgs2, h=0.1
ic4 vs bsgs2, h=1
0.5
1.0
1.5
2.0
2.5
3.0
22b T=1024
22b T=4096
22b T=16384
24b T=1024
24b T=4096
24b T=16384
28b T=1024
28b T=4096
28b T=16384
32b T=1024
32b T=4096
32b T=16384